Final Project (6 marks)

Machine Learning 2026 – LIACD, LECD, LEEC
Informatics Engineering Department

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1. Problem Statement

In this assignment, you must apply the methods you have learned throughout the semester to a stress level classification problem, whose data set is available for download on the Kaggle platform at the following link:

https://www.kaggle.com/datasets/jayjoshi37/sleep-screen-time-and-stress-analysis

The goal of this project is to tackle three problems:

- **A.** Predict the stress level in a binary setting.
- **B.** Predict the stress level in a 5-class setting.
- **C.** Predict the stress level in a 10-class setting, from no or low to maximum stress.

2. Dataset Description (adapted from Kaggle)

As mobile devices become inseparable from daily life, understanding their impact on mental well-being is critical. The 'Sleep, Screen Time and Stress' data set explores the relationship between smartphone usage, sleep patterns, and stress levels, featuring 15,000 synthetic records detailing the lifestyles and mobile habits of a diverse population. By tracking metrics such as pre-sleep screen time, physical activity levels, and caffeine intake, this data can be modeled to classify stress.

3. Objectives

This project is divided into the following three modelling scenarios:

- **3.1. Scenario A (Binary Problem)** – The objective of this scenario is to classify each instance as, for example, 'no stress' (0) or 'stress' (1);
- **3.2. Scenario B (Multi-Class Problem 1)** – The objective of this scenario is to classify the level of stress as one of five possible levels, e.g. , 'no stress' (0), 'low' (1), 'moderate' (2), 'high' (3) and 'extreme' (4);
- **3.3. Scenario C (Multi-Class Problem 2)** – The objective of this scenario is to classify the level of stress as one of 10 possible, e.g., from 'no stress' (0) to 'maximum' (9).

Two important notes:

1. The stress level is originally recorded as a continuous scale. Make the necessary arrangements to transform the associated variable to match each scenarios' requirements. For example, in the binary case, you can set the 'no stress'/'low stress' class to encompass all the values in [0, 5[, whereas the 'stress' will contain all within [5, 10] (equal-width binning). Other discretization methods can be used, such as equal-frequency binning or K-means clustering to identify natural groupings, depending on the variable's distribution.

2. The class nature and names are flexible. You can change them to reflect your own preference.

4. Practical Assignment

4.1. Data Import

Develop scripts for feature data import. Organize data into sub-sets relating to each source type you intend to test, e.g., create training, validation, and testing sets. If present, remove or find ways to handle missing data (for example, doing the mean or the median of a specific feature), check for duplicates, inconsistencies, etc. In sum, do a general data quality check before processing with the next steps.

4.2. Data Analysis

You need to understand the data you are working with. To that end, you should explore it. Use different data visualization tools (histograms, pie charts, box plots, correlation plots) to analyze the data. Consider using feature selection technics and see how they affect the performance of the machine learning algorithms. Notice the class imbalance and the different features' domains in this dataset and find strategies to cope. Make sure you know your features.

Do any pre-processing, data encoding, or normalization that you deem necessary, always justifying your decisions. **Remember to include in the final report all your findings, challenges, and steps taken.**

4.3. Experimental Analysis

You should be able to design experiences, run machine learning algorithms on the given data, and evaluate their results. Define the appropriate performance metrics and justify your choices. In your report, you can also include a schema of your experimental pipeline.

Run the experiments multiple times. You should present average results and standard deviations (of the metrics used). You may also decide to cross-validate. Ultimately, you should be able to choose the best classifier and evaluate them in a testing set (hold out).

Do not forget that manually inspecting the predictions of your algorithms can give you precious insights into where they might be failing (and why) and what you can do to improve them (e.g., what makes the algorithm fail in this particular case? what special characteristic does it have that makes it so hard? how can I make the algorithm better deal with those cases?). Go back and forward to the hyperparameter choice of the different machine learning algorithms until you are satisfied with the results. It is a good idea to keep track of the evolution of the performance of your algorithm during this process. Try to show these trends in your final report to justify all the involved issues (choosing parameters, model fit, etc.). You should try to understand how they perform differently in your data.

With respect to the models, you must use **3 different machine learning methods**: Support Vector Machines (SVM) and Neural Networks (NN), which are **mandatory**, and are under severe grade penalty if not used. The third method is up to you to choose, except for KNN and simple Decision Trees used in the Challenge. Justify your choice.

4.4. Libraries and Language

Your code must be written in Python. You are free to use Colab, Jupyter, or create a script. For Neural Networks, we recommend you use PyTorch, Keras, Tensorflow, or Scikit-learn (for more basic architectures). In any case, you are to use any Python-based framework/library.

4.5. Results and Discussion

Present and discuss the results obtained from your experiments. Other authors have already studied this type of problem and related data sets. Thus, **compare** yours with the results from other sources, and **do not copy** from them. **Do not forget to cite your sources.**

5. Documentation

5.1. Description

Write documentation (in Portuguese or English) about your project. The documentation should include a cover page where the course name, project title, date, names, and student numbers of the authors are mentioned.

Describe the methods used for classification in such detail that the reader would be able to implement the same functions for feature extraction and classification based on your documentation and some basic background in pattern recognition. Always justify your choices, even when they are based on intuition. Do not forget to verify your assumptions! Include classification results with the given data in your documentation. At the end of your documentation, you should have a list of all references used.

5.2. Report Structure

The report must be delivered in PDF format. A recommended report structure can include the following sections:

- **Cover Page**: Course name, project title, date, authors, emails, student numbers, and eventually a graphical abstract.
- **Introduction**: Problem presentation and summary of your methodology and key findings.
- **Background**: Any relevant information about your methodology or other works used to compare your methodology.
- **Exploratory Data Analysis**: Summary statistics (e.g., mean, median, standard deviation), data cleaning process (handling missing values, outliers, etc.), or visualizations such as histograms, scatter plots, correlation matrices, or box plots.
- **Methodologies**: Explain explains the step-by-step approach taken to analyze the data and solve the problem, description of techniques and models used, justification for selecting specific methodologies, explanation, and visualization of the work pipeline (e.g., preprocessing, feature selection, model training, etc.).
- **Results and Discussion**: Presentation of key results (e.g., tables, graphs, performance metrics), interpretation of results and their implications, comparison with baseline models or other methodologies, discussion of any limitations or challenges encountered. All conducted experiments should be reported and all decisions justified.
- **Conclusions**: Recap of major findings, potential future work, or improvements.

- **References**: Any reference used in your report, including the dataset itself.

5.3. Requirements, Submission, Discussion

Similar to the Challenge, this assignment is meant to be done in groups of two. If someone wants to work alone, although not advisable, it is also possible. Larger groups are not allowed without proper justification and permission.

Final Project Deadline: **May 21st, 2026.**

Deliverables:

- Final report (in PDF).
- Source code (.py or .ipynb files).

Project discussion: **June 3rd**.

NOTE

The use of AI tools and technologies, including Generative AI tools (e.g., ChatGPT, Claude, Gemini, or similar LLMs), to assist with coding and writing is permitted. However, to maintain transparency, authors must fully and honestly disclose any utilization of AI-assisted tools in their work, particularly generative AI technologies (e.g., ChatGPT, Claude, Gemini, or similar LLMs). Disclosures should encompass both writing and coding contributions. For instance, authors might include statements in the Acknowledgements/Disclaiming section at the end of the paper such as: "ChatGPT was used to generate a draft of Section X," or "Claude was utilized to polish English of paragraph Y on page Z." Additionally, statements like "Gemini was employed to produce tables, graphs, code, data, citations, etc." can provide clarity on the specific role of AI tools.